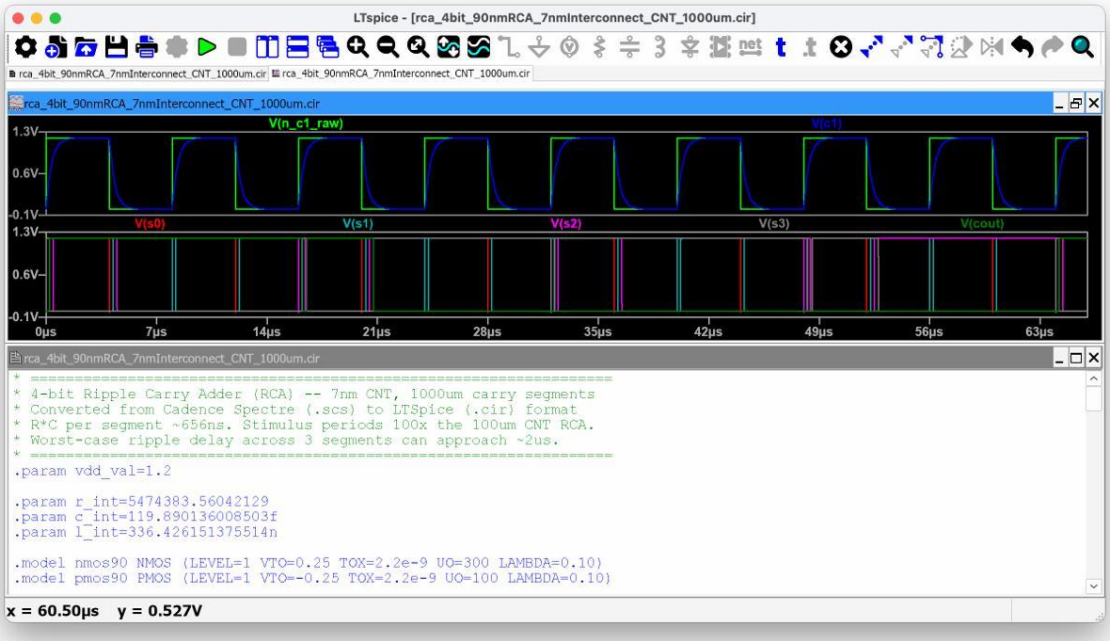
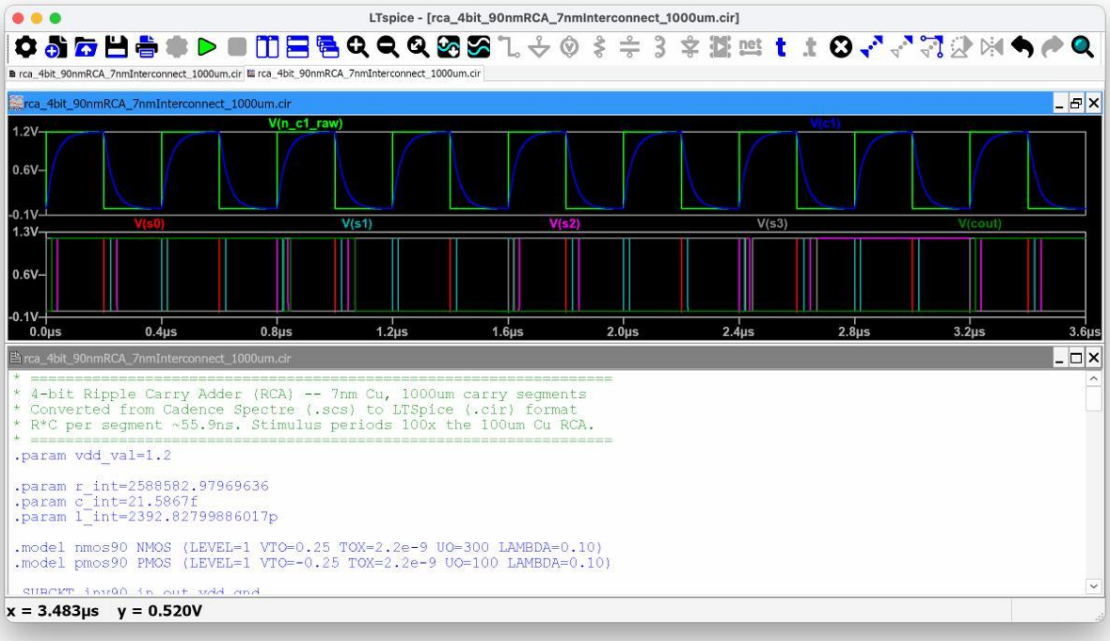
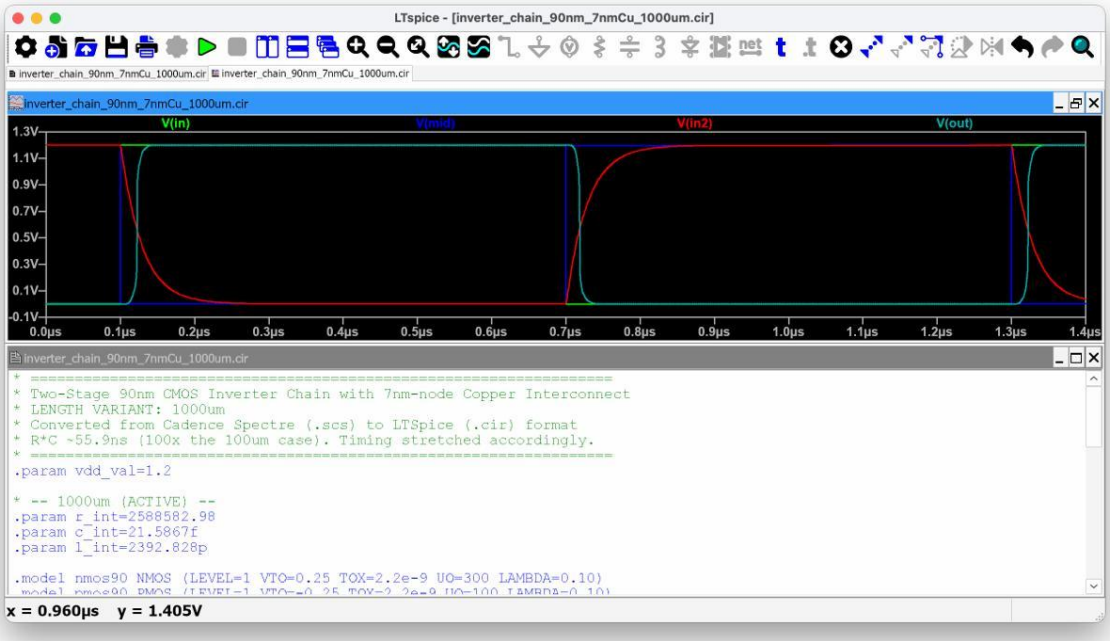
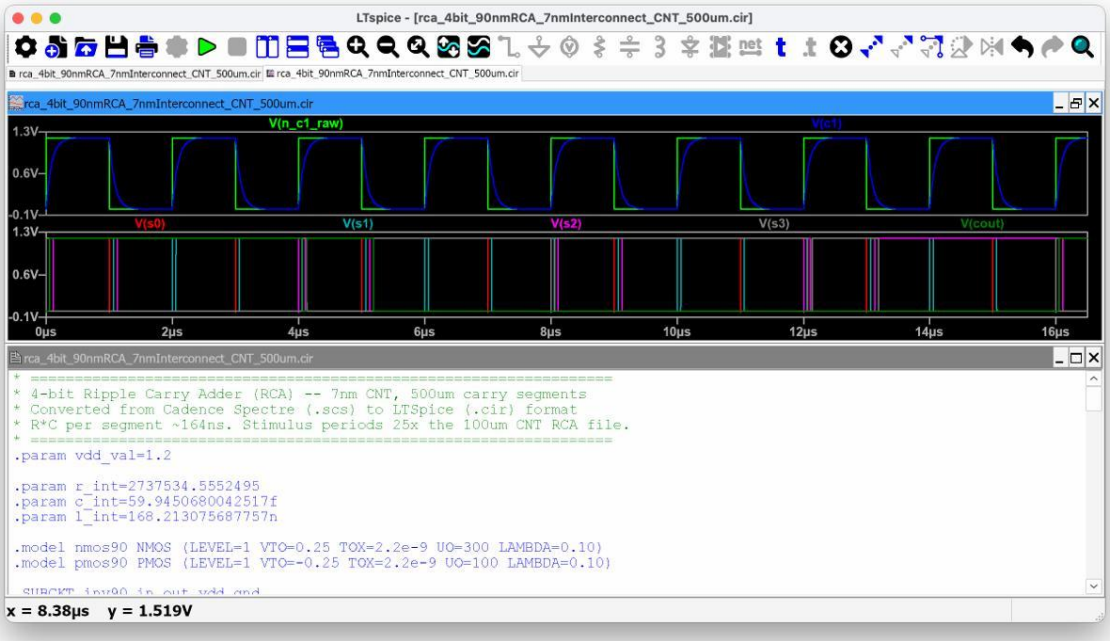
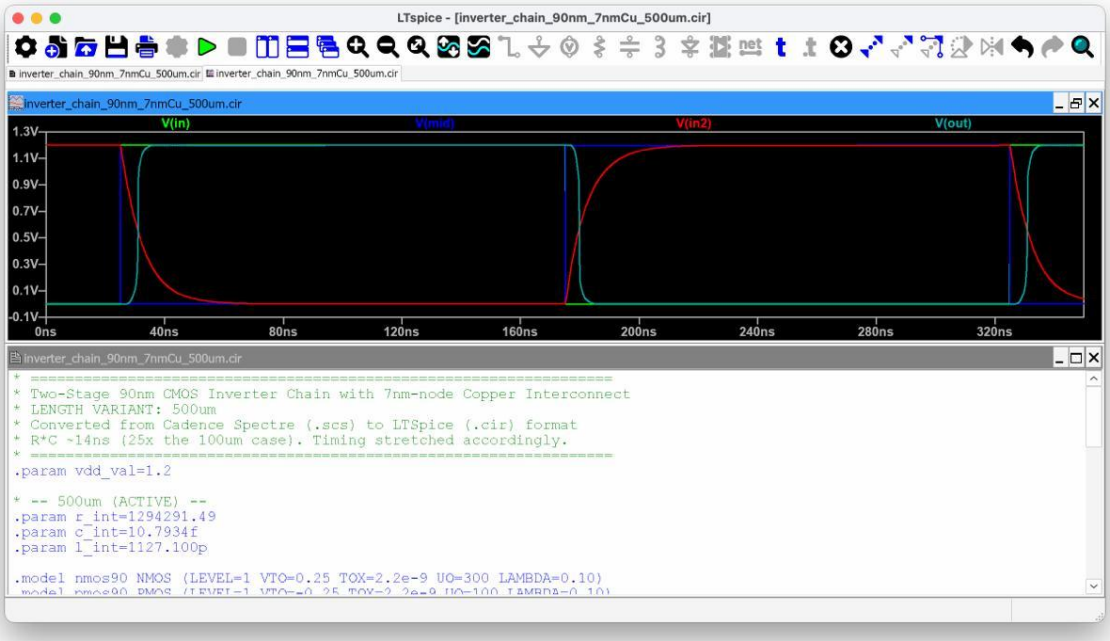
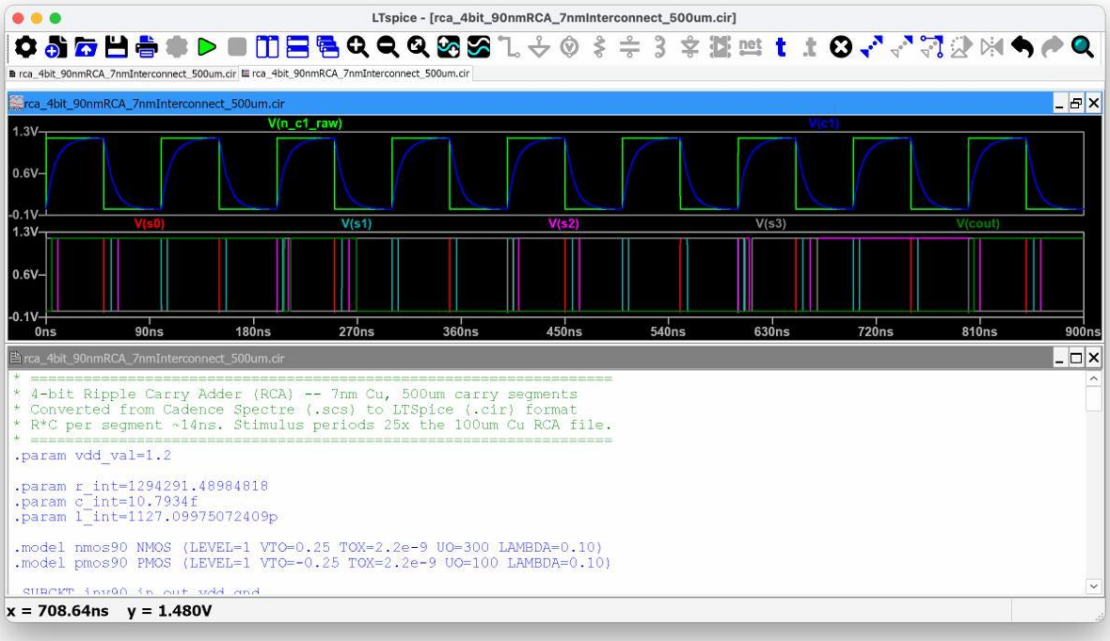

LTspice - [rca_4bit_90nmRCA_7nmInterconnect_CNT_1000um.cir]
rca_4bit_90nmRCA_7nmInterconnect_CNT_1000um.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
0µs
7µs
14µs
21µs
28µs
35µs
42µs
49µs
56µs
63µs
* ==========
* 4-bit Ripple Carry Adder (RCA) -- 7nm CNT, 1000um carry segments
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C per segment ~656ns. Stimulus periods 100x the 100um CNT RCA.
* Worst-case ripple delay across 3 segments can approach ~2us.
* ==========
.param vdd_val=1.2
.param r_int=5474383.56042129
.param c_int=119.890136008503f
.param l_int=336.426151375514n
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
.model pmos90 PMOS (LEVEL=1 VTO=-0.25 TOX=2.2e-9 UO=100 LAMBDA=0.10)
x = 60.50µs y = 0.527V

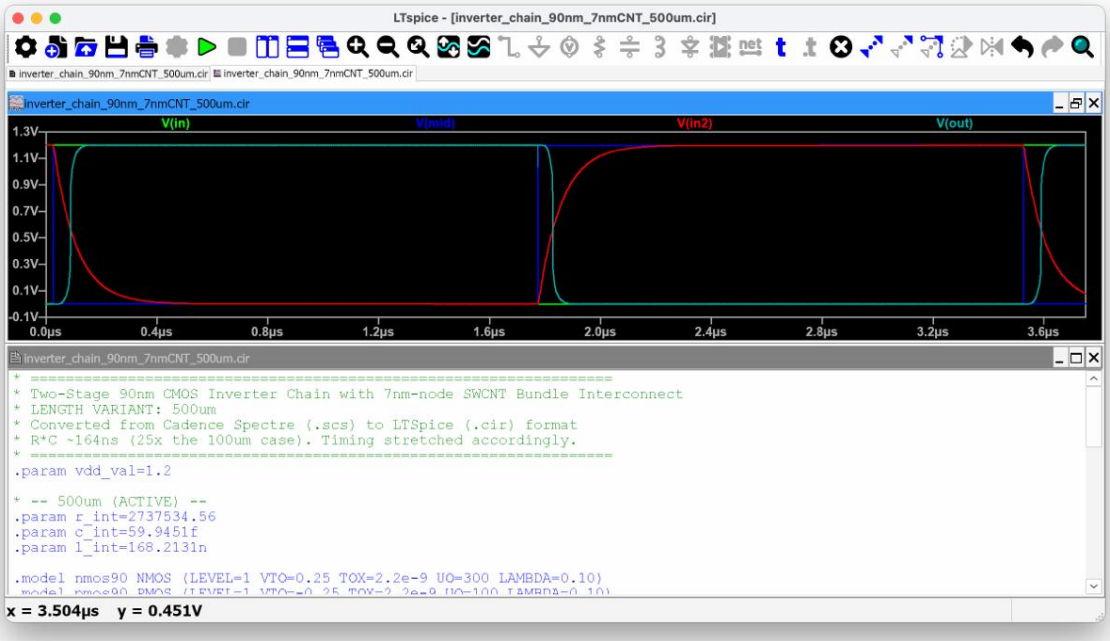

LTspice - [rca_4bit_90nmRCA_7nmInterconnect_1000um.cir]
rca_4bit_90nmRCA_7nmInterconnect_1000um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0.0µs
0.4µs
0.8µs
1.2µs
1.6µs
2.0µs
2.4µs
2.8µs
3.2µs
3.6µs
* ==========
* 4-bit Ripple Carry Adder (RCA) -- 7nm Cu, 1000um carry segments
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C per segment ~55.9ns. Stimulus periods 100x the 100um Cu RCA.
* ==========
.param vdd_val=1.2
.param r_int=2588582.97969636
.param c_int=21.5867f
.param l_int=2392.82799886017p
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
.model pmos90 PMOS (LEVEL=1 VTO=-0.25 TOX=2.2e-9 UO=100 LAMBDA=0.10)
x = 3.483µs y = 0.520V

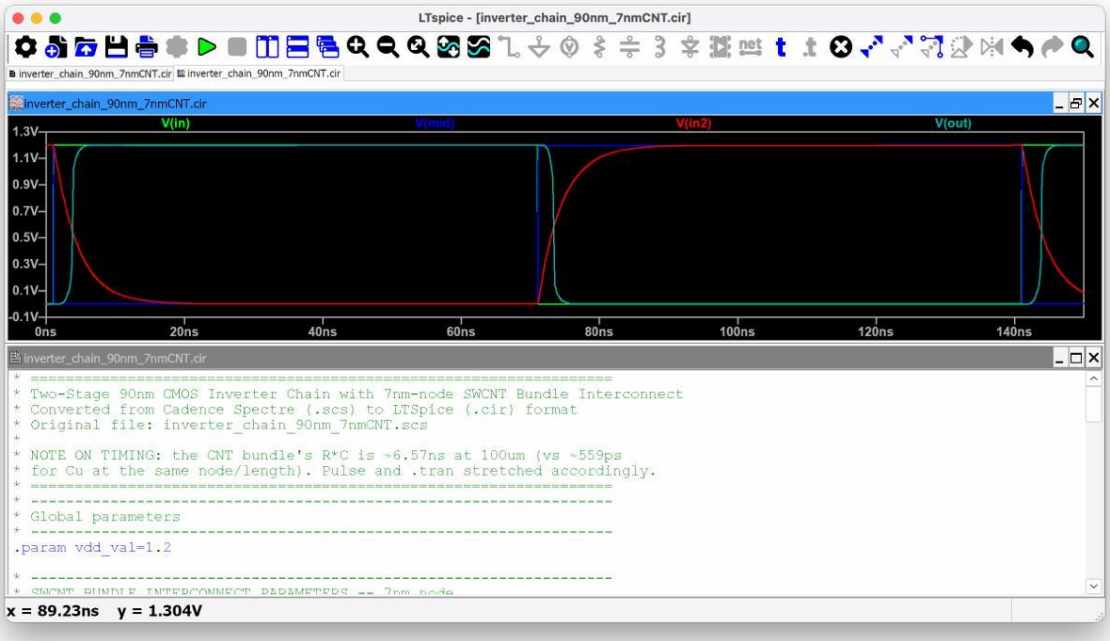

LTspice - [inverter_chain_90nm_7nmCu_1000um.cir]
inverter_chain_90nm_7nmCu_1000um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0µs
0.1µs
0.2µs
0.3µs
0.4µs
0.5µs
0.6µs
0.7µs
0.8µs
0.9µs
1.0µs
1.1µs
1.2µs
1.3µs
1.4µs
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 7nm-node Copper Interconnect
* LENGTH VARIANT: 1000um
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C ~55.9ns (100x the 100um case). Timing stretched accordingly.
* ==========
.param vdd_val=1.2
* -- 1000um (ACTIVE) --
.param r_int=2588582.98
.param c_int=21.5867f
.param l_int=2392.828p
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
x = 0.960µs y = 1.405V

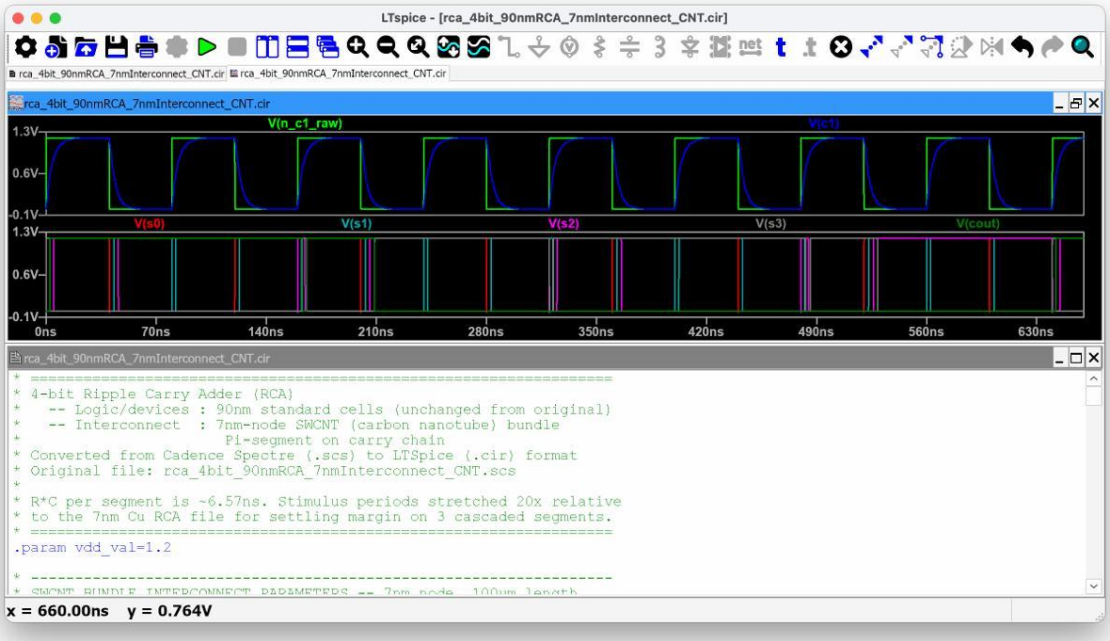

LTspice - [rca_4bit_90nmRCA_7nmInterconnect_CNT_500um.cir]
rca_4bit_90nmRCA_7nmInterconnect_CNT_500um.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0µs
2µs
4µs
6µs
8µs
10µs
12µs
14µs
16µs
* ==========
* 4-bit Ripple Carry Adder (RCA) -- 7nm CNT, 500um carry segments
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C per segment ~164ns. Stimulus periods 25x the 100um CNT RCA file.
* ==========
.param vdd_val=1.2
.param r_int=2737534.5552495
.param c_int=59.94506800425l7f
.param l_int=168.213075687757n
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
.model pmos90 PMOS (LEVEL=1 VTO=-0.25 TOX=2.2e-9 UO=100 LAMBDA=0.10)
x = 8.38µs y = 1.519V

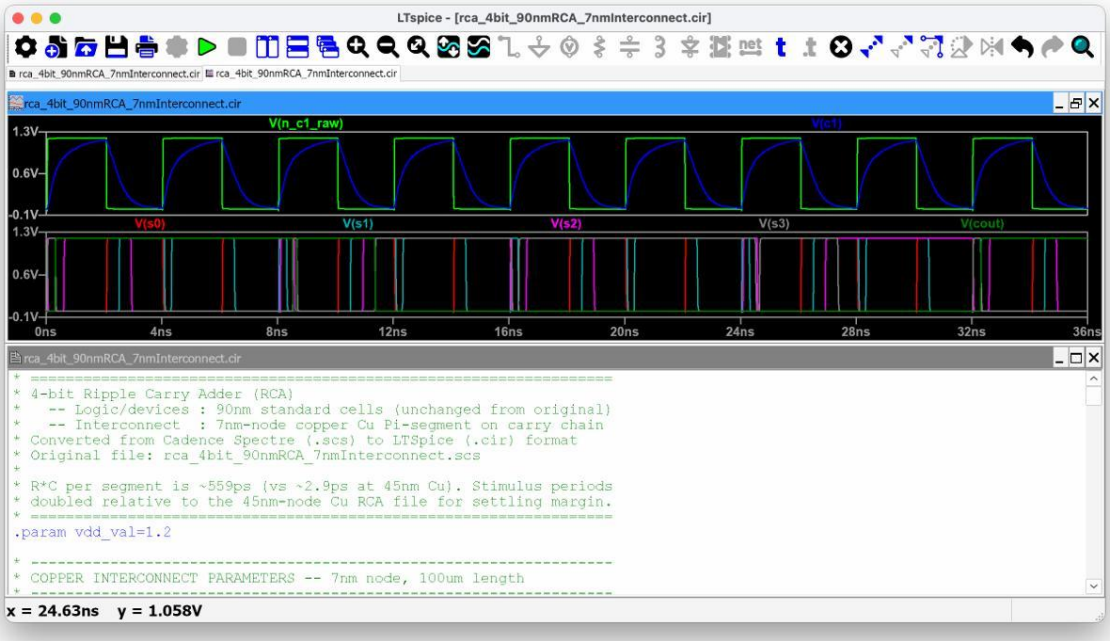

LTspice - [inverter_chain_90nm_7nmCu_500um.cir]
inverter_chain_90nm_7nmCu_500um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
40ns
80ns
120ns
160ns
200ns
240ns
280ns
320ns
inverter_chain_90nm_7nmCu_500um.cir
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 7nm-node Copper Interconnect
* LENGTH VARIANT: 500um
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C ~14ns (25x the 100um case). Timing stretched accordingly.
* ==========
.param vdd_val=1.2
* -- 500um (ACTIVE) --
.param r_int=1294291.49
.param c_int=10.7934f
.param l_int=1127.100p
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)

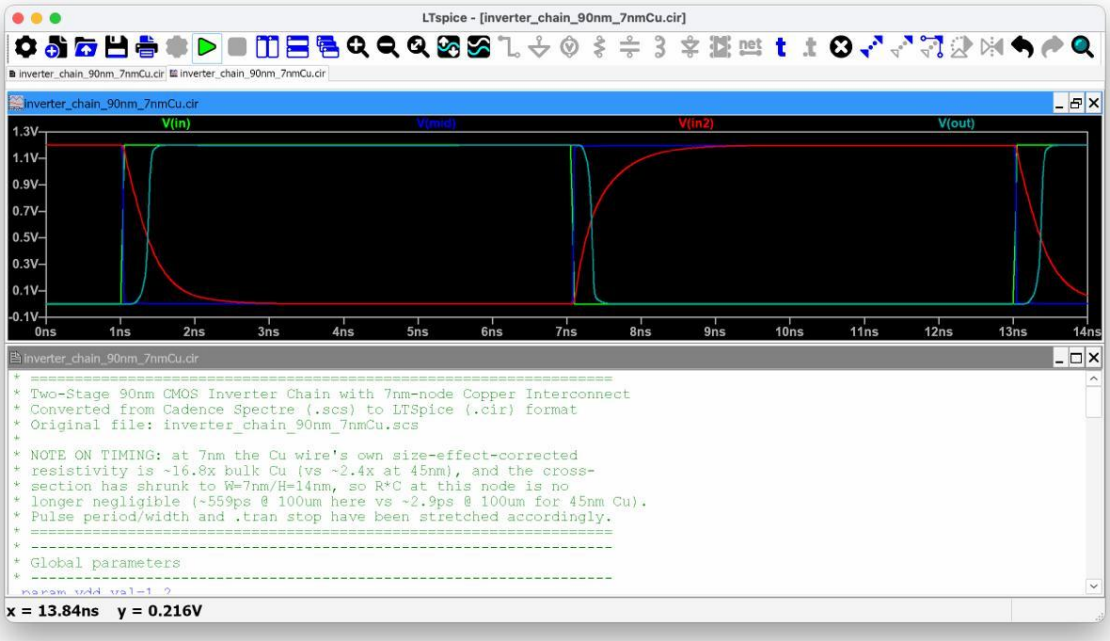

LTspice - [rca_4bit_90nmRCA_7nmInterconnect_500um.cir]
rca_4bit_90nmRCA_7nmInterconnect_500um.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
0ns
90ns
180ns
270ns
360ns
450ns
540ns
630ns
720ns
810ns
900ns
rca_4bit_90nmRCA_7nmInterconnect_500um.cir
* ==========
* 4-bit Ripple Carry Adder (RCA) -- 7nm Cu, 500um carry segments
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C per segment ~14ns. Stimulus periods 25x the 100um Cu RCA file.
* ==========
.param vdd_val=1.2
.param r_int=1294291.48984818
.param c_int=10.7934f
.param l_int=1127.09975072409p
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
.model pmos90 PMOS (LEVEL=1 VTO=-0.25 TOX=2.2e-9 UO=100 LAMBDA=0.10)
x = 708.64ns y = 1.480V


LTspice - [inverter_chain_90nm_7nmCNT_500um.cir]
inverter_chain_90nm_7nmCNT_500um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0µs
0.4µs
0.8µs
1.2µs
1.6µs
2.0µs
2.4µs
2.8µs
3.2µs
3.6µs
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 7nm-node SWCNT Bundle Interconnect
* LENGTH VARIANT: 500um
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C ~164ns (25x the 100um case). Timing stretched accordingly.
* ==========
.param vdd_val=1.2
* -- 500um (ACTIVE) --
.param r_int=2737534.56
.param c_int=59.9451f
.param l_int=168.2131n
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
x = 3.504µs y = 0.451V

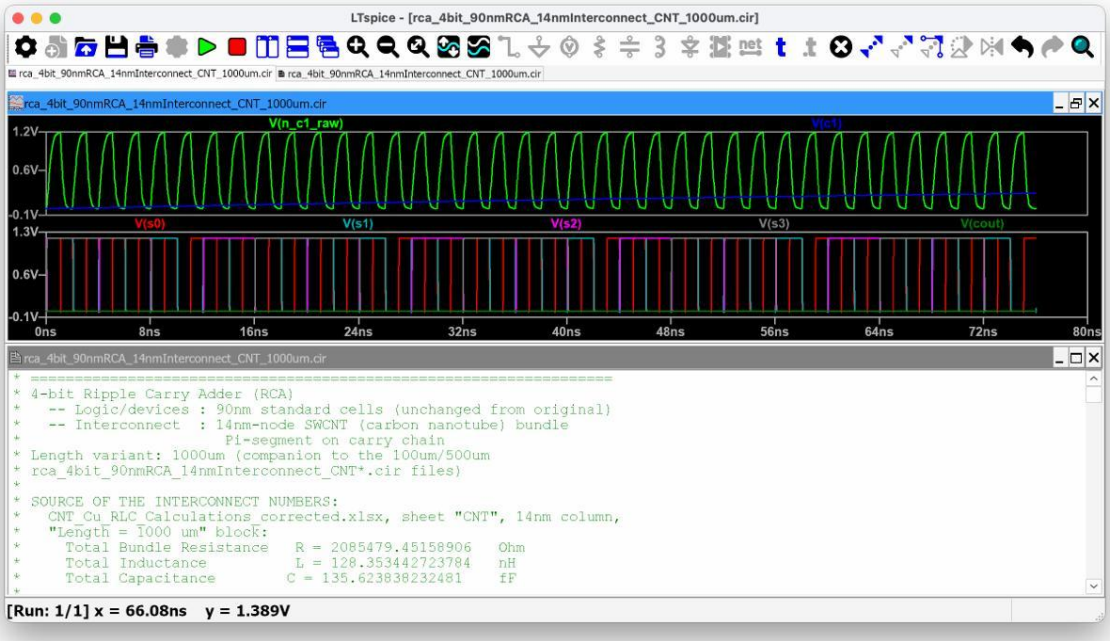

LTspice - [inverter_chain_90nm_7nmCNT.cir]
inverter_chain_90nm_7nmCNT.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
20ns
40ns
60ns
80ns
100ns
120ns
140ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 7nm-node SWCNT Bundle Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_7nmCNT.scs
*
* NOTE ON TIMING: the CNT bundle's R*C is ~6.57ns at 100um (vs ~559ps
* for Cu at the same node/length). Pulse and .tran stretched accordingly.
* ==========
* ----------
* Global parameters
* ----------
.param vdd_val=1.2
* ----------
x = 89.23ns y = 1.304V

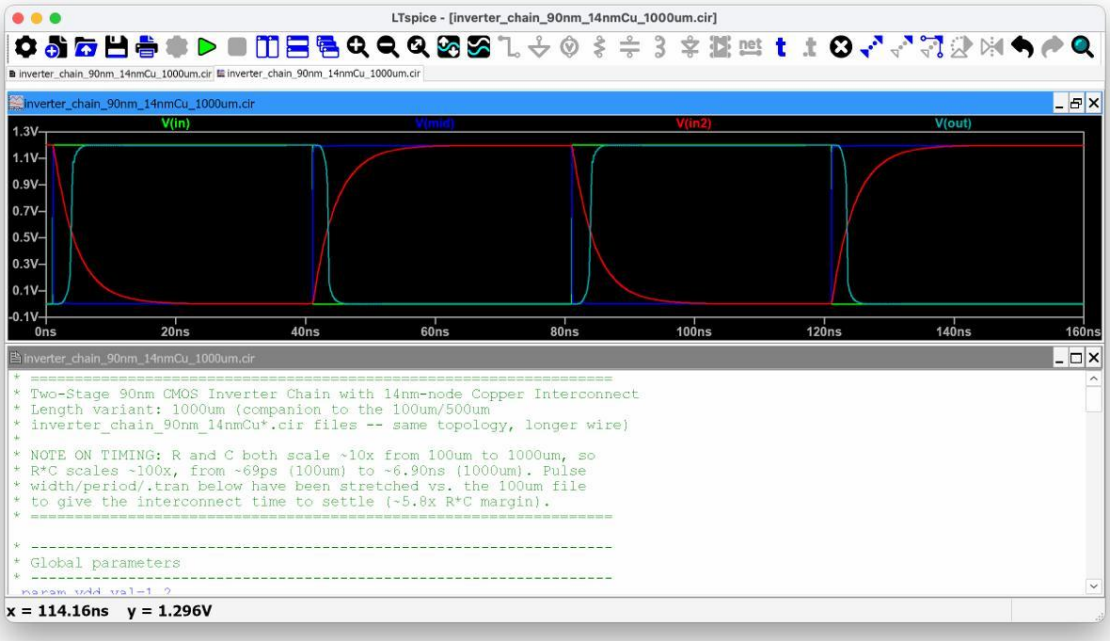

LTspice - [rca_4bit_90nmRCA_7nmInterconnect_CNT.cir]
rca_4bit_90nmRCA_7nmInterconnect_CNT.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
0ns
70ns
140ns
210ns
280ns
350ns
420ns
490ns
560ns
630ns
* ====================================================================
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 7nm-node SWCNT (carbon nanotube) bundle
*                      Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_7nmInterconnect_CNT.scs
*
* R*C per segment is ~6.57ns. Stimulus periods stretched 20x relative
* to the 7nm Cu RCA file for settling margin on 3 cascaded segments.
* ====================================================================
.param vdd_val=1.2
x = 660.00ns  y = 0.764V

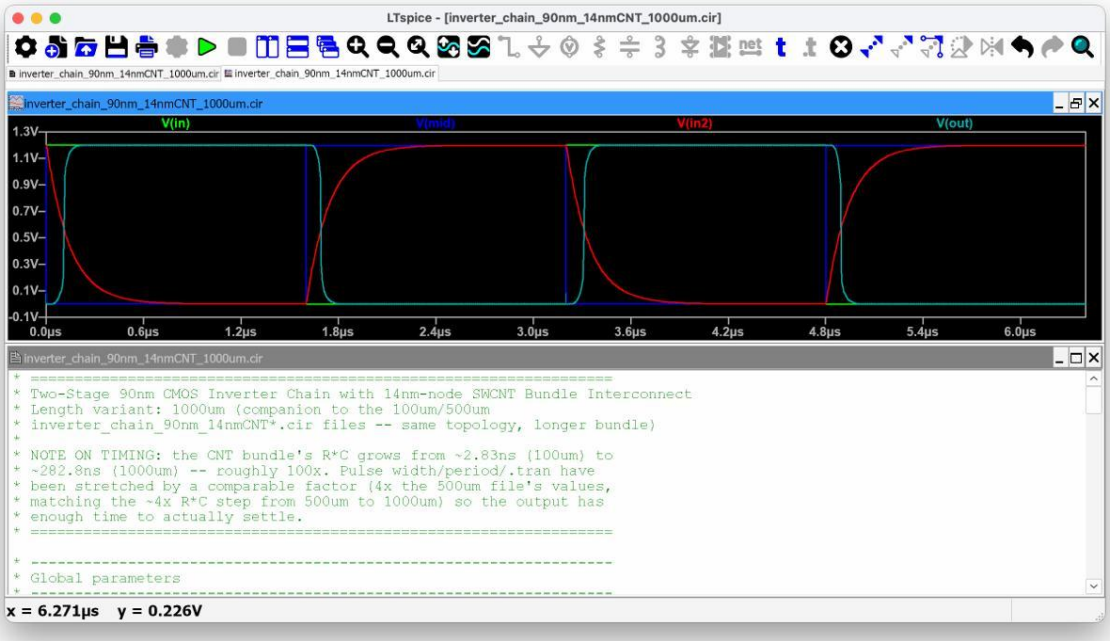

LTspice - [rca_4bit_90nmRCA_7nmInterconnect.cir]
rca_4bit_90nmRCA_7nmInterconnect.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
0ns
4ns
8ns
12ns
16ns
20ns
24ns
28ns
32ns
36ns
* ====================================================================
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 7nm-node copper Cu Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_7nmInterconnect.scs
*
* R*C per segment is ~559ps (vs ~2.9ps at 45nm Cu). Stimulus periods
* doubled relative to the 45nm-node Cu RCA file for settling margin.
* ====================================================================
.param vdd_val=1.2
* ------------------------------------------------------------------
* COPPER INTERCONNECT PARAMETERS -- 7nm node, 100um length
* ------------------------------------------------------------------
x = 24.63ns  y = 1.058V

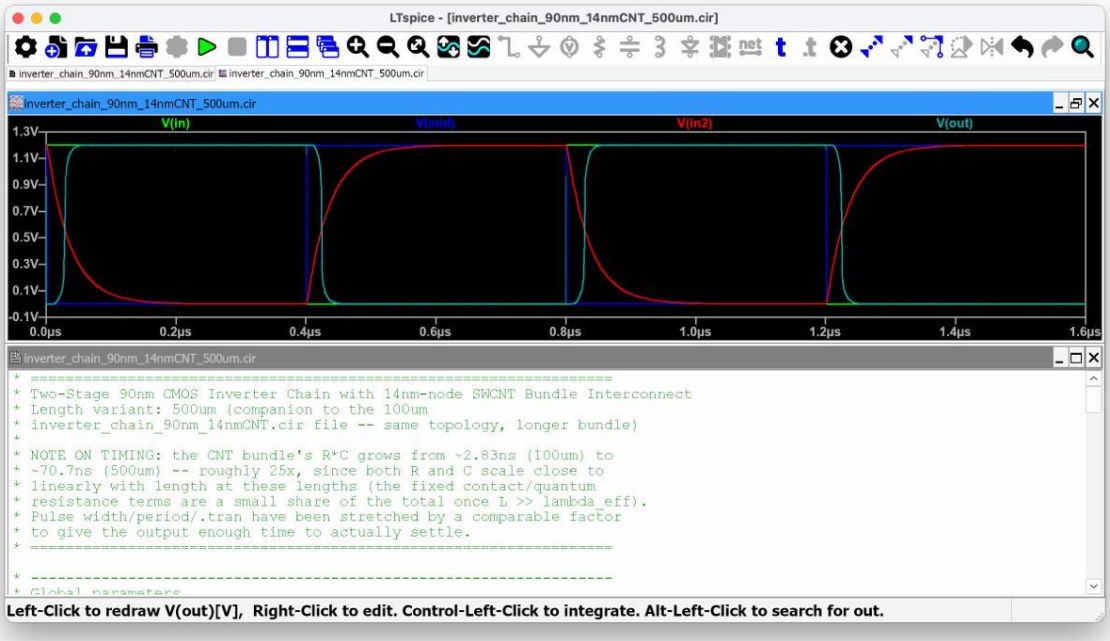

LTspice - [inverter_chain_90nm_7nmCu.cir]
inverter_chain_90nm_7nmCu.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
1ns
2ns
3ns
4ns
5ns
6ns
7ns
8ns
9ns
10ns
11ns
12ns
13ns
14ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 7nm-node Copper Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_7nmCu.scs
*
* NOTE ON TIMING: at 7nm the Cu wire's own size-effect-corrected
* resistivity is ~16.8x bulk Cu (vs ~2.4x at 45nm), and the cross-
* section has shrunk to W=7nm/H=14nm, so R*C at this node is no
* longer negligible (~559ps @ 100um here vs ~2.9ps @ 100um for 45nm Cu).
* Pulse period/width and .tran stop have been stretched accordingly.
* ==========
* ----------
* Global parameters
* ----------
x = 13.84ns y = 0.216V

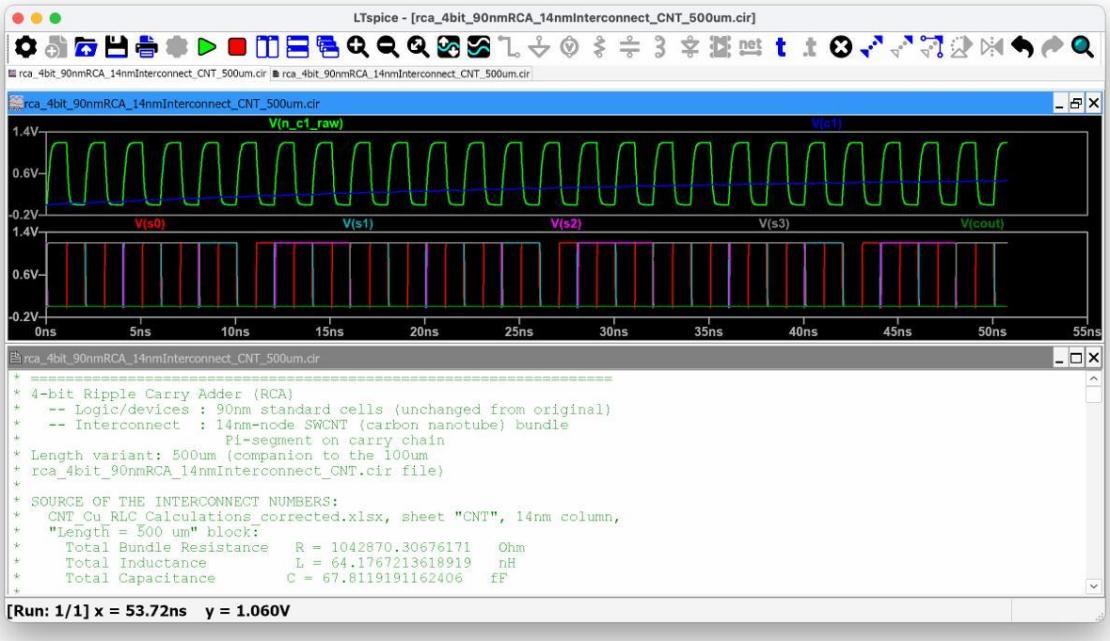

LTspice - [rca_4bit_90nmRCA_14nmInterconnect_1000um.cir]
rca_4bit_90nmRCA_14nmInterconnect_1000um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
18ns
20ns
22ns
24ns
26ns
* ==========
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 14nm-node copper Cu Pi-segment on carry chain
* Length variant: 1000um (companion to the 100um/500um
* rca_4bit_90nmRCA_14nmInterconnect*.cir files)
*
* SOURCE OF THE INTERCONNECT NUMBERS:
*   CNT_Cu_RLC_Calculations_corrected.xlsx, sheet "Cu", 14nm column,
*   "Length = 1000 um" block:
*     Total Resistance      R = 280056.517711655   Ohm
*     Total Self-Inductance L = 2.26097974920331   nH
*     Total Capacitance     C = 24.6436011096596   fF
*
Ready

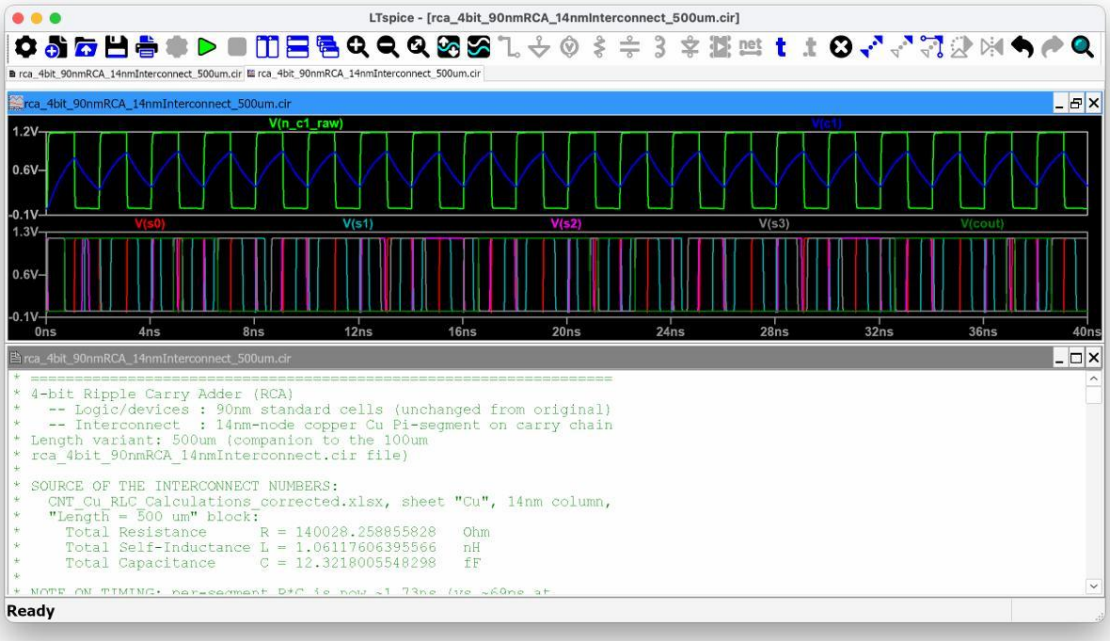

LTspice - [rca_4bit_90nmRCA_14nmInterconnect_CNT_1000um.cir]
rca_4bit_90nmRCA_14nmInterconnect_CNT_1000um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
8ns
16ns
24ns
32ns
40ns
48ns
56ns
64ns
72ns
80ns
* ==========
* 4-bit Ripple Carry Adder (RCA)
* -- Logic/devices : 90nm standard cells (unchanged from original)
* -- Interconnect : 14nm-node SWCNT (carbon nanotube) bundle
* Pi-segment on carry chain
* Length variant: 1000um (companion to the 100um/500um
* rca_4bit_90nmRCA_14nmInterconnect_CNT*.cir files)
*
* SOURCE OF THE INTERCONNECT NUMBERS:
* CNT_Cu_RLC_Calculations_corrected.xlsx, sheet "CNT", 14nm column,
* "Length = 1000 um" block:
* Total Bundle Resistance R = 2085479.45158906 Ohm
* Total Inductance L = 128.353442723784 nH
* Total Capacitance C = 135.623838232481 fF
*
[Run: 1/1] x = 66.08ns y = 1.389V

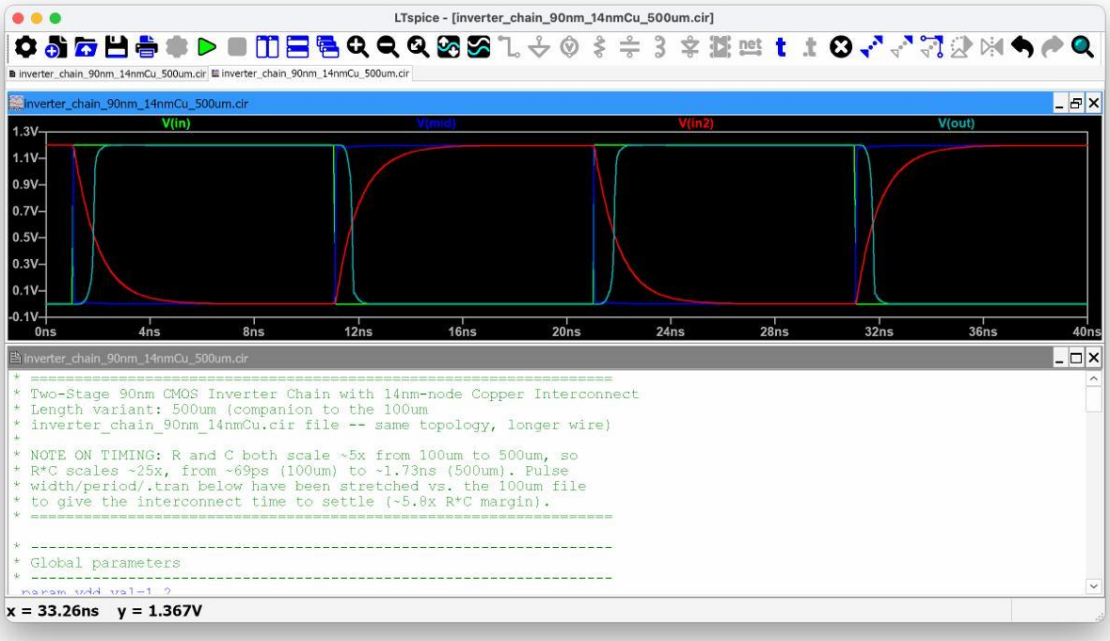

LTspice - [inverter_chain_90nm_14nmCu_1000um.cir]
inverter_chain_90nm_14nmCu_1000um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
20ns
40ns
60ns
80ns
100ns
120ns
140ns
160ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 14nm-node Copper Interconnect
* Length variant: 1000um (companion to the 100um/500um
* inverter_chain_90nm_14nmCu*.cir files -- same topology, longer wire)
*
* NOTE ON TIMING: R and C both scale ~10x from 100um to 1000um, so
* R*C scales ~100x, from ~69ps (100um) to ~6.90ns (1000um). Pulse
* width/period/.tran below have been stretched vs. the 100um file
* to give the interconnect time to settle (~5.8x R*C margin).
* ==========
* ----------
* Global parameters
* ----------
.param vdd val=1.2
x = 114.16ns y = 1.296V


LTspice - [inverter_chain_90nm_14nmCNT_1000um.cir]
inverter_chain_90nm_14nmCNT_1000um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0µs
0.6µs
1.2µs
1.8µs
2.4µs
3.0µs
3.6µs
4.2µs
4.8µs
5.4µs
6.0µs
* Two-Stage 90nm CMOS Inverter Chain with 14nm-node SWCNT Bundle Interconnect
* Length variant: 1000um (companion to the 100um/500um
* inverter_chain_90nm_14nmCNT*.cir files -- same topology, longer bundle)
* NOTE ON TIMING: the CNT bundle's R*C grows from ~2.83ns (100um) to
* ~282.8ns (1000um) -- roughly 100x. Pulse width/period/.tran have
* been stretched by a comparable factor (4x the 500um file's values,
* matching the ~4x R*C step from 500um to 1000um) so the output has
* enough time to actually settle.
* Global parameters
x = 6.271µs y = 0.226V

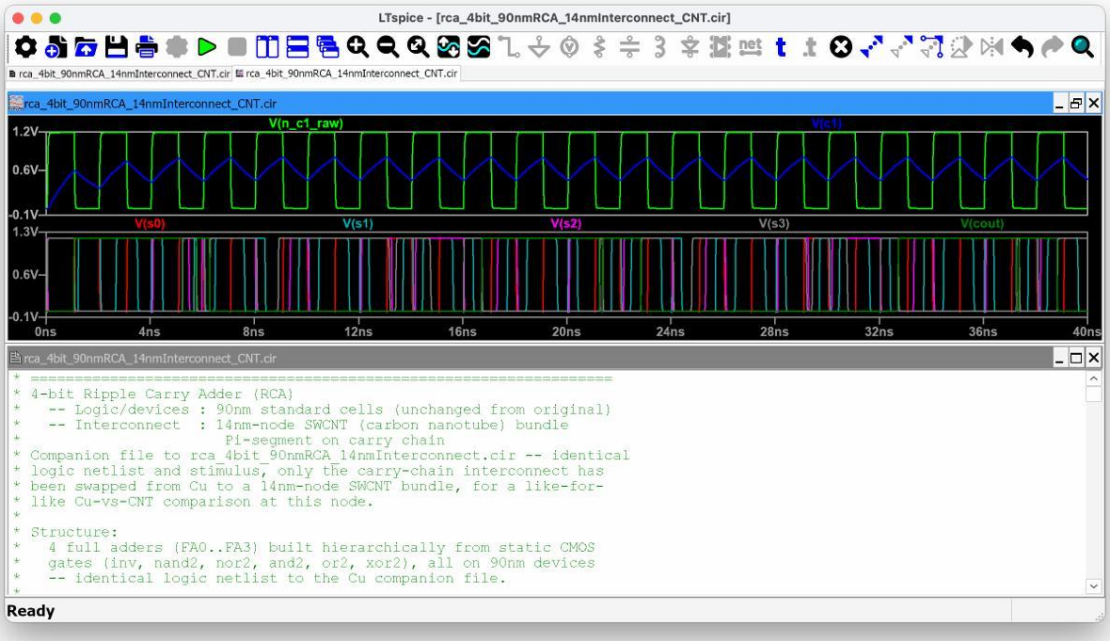

LTspice - [inverter_chain_90nm_14nmCNT_500um.cir]
inverter_chain_90nm_14nmCNT_500um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0µs
0.2µs
0.4µs
0.6µs
0.8µs
1.0µs
1.2µs
1.4µs
1.6µs
* Two-Stage 90nm CMOS Inverter Chain with 14nm-node SWCNT Bundle Interconnect
* Length variant: 500um (companion to the 100um
* inverter_chain_90nm_14nmCNT.cir file -- same topology, longer bundle)
* NOTE ON TIMING: the CNT bundle's R*C grows from ~2.83ns (100um) to
* ~70.7ns (500um) -- roughly 25x, since both R and C scale close to
* linearly with length at these lengths (the fixed contact/quantum
* resistance terms are a small share of the total once L >> lambda_eff).
* Pulse width/period/.tran have been stretched by a comparable factor
* to give the output enough time to actually settle.
Left-Click to redraw V(out)[V], Right-Click to edit. Control-Left-Click to integrate. Alt-Left-Click to search for out.

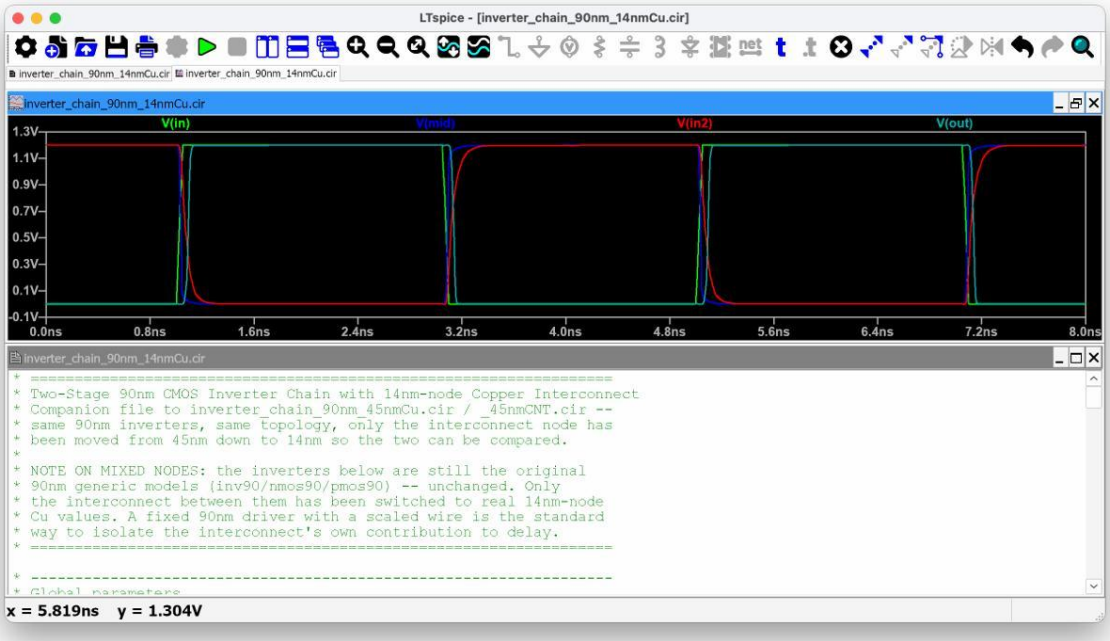

LTspice - [rca_4bit_90nmRCA_14nmInterconnect_CNT_500um.cir]
rca_4bit_90nmRCA_14nmInterconnect_CNT_500um.cir
V(n_c1_raw)
V(c1)
1.4V
0.6V
-0.2V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
0ns
5ns
10ns
15ns
20ns
25ns
30ns
35ns
40ns
45ns
50ns
55ns
* ====================================================================
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 14nm-node SWCNT (carbon nanotube) bundle
*                      Pi-segment on carry chain
* Length variant: 500um (companion to the 100um
* rca_4bit_90nmRCA_14nmInterconnect_CNT.cir file)
*
* SOURCE OF THE INTERCONNECT NUMBERS:
*   CNT_Cu_RLC_Calculations_corrected.xlsx, sheet "CNT", 14nm column,
*   "Length = 500 um" block:
*     Total Bundle Resistance   R = 1042870.30676171   Ohm
*     Total Inductance          L = 64.1767213618919   nH
*     Total Capacitance         C = 67.8119191162406   fF
*
[Run: 1/1] x = 53.72ns  y = 1.060V

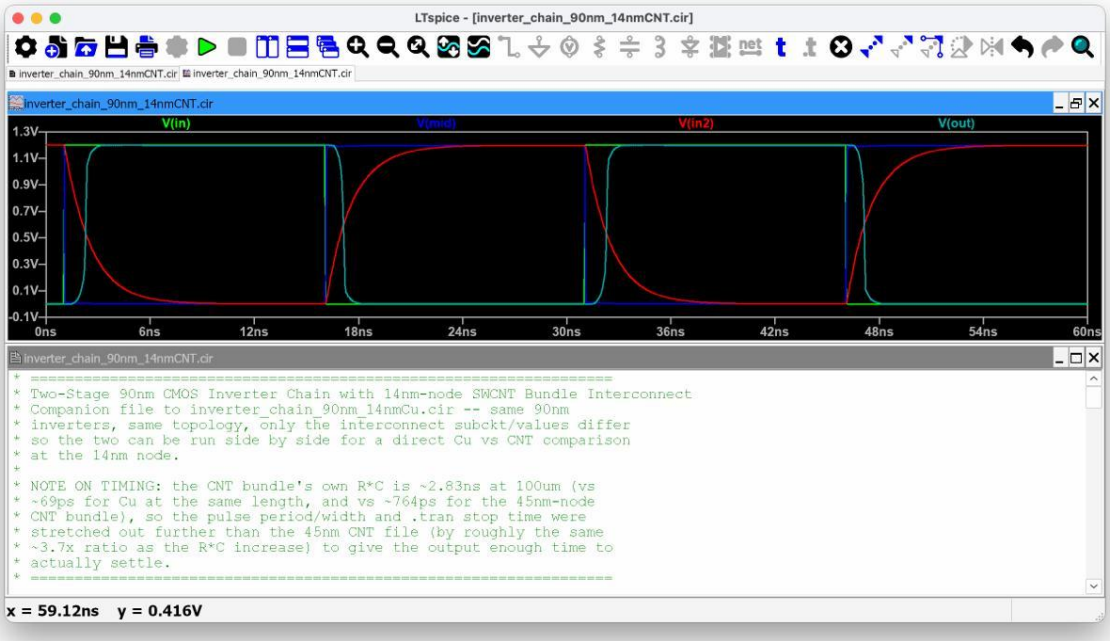

LTspice - [rca_4bit_90nmRCA_14nmInterconnect_500um.cir]
rca_4bit_90nmRCA_14nmInterconnect_500um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
4ns
8ns
12ns
16ns
20ns
24ns
28ns
32ns
36ns
40ns
* ====================================================================
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 14nm-node copper Cu Pi-segment on carry chain
* Length variant: 500um (companion to the 100um
* rca_4bit_90nmRCA_14nmInterconnect.cir file)
*
* SOURCE OF THE INTERCONNECT NUMBERS:
*   CNT_Cu_RLC_Calculations_corrected.xlsx, sheet "Cu", 14nm column,
*   "Length = 500 um" block:
*     Total Resistance      R = 140028.258855828   Ohm
*     Total Self-Inductance L = 1.06117606395566   nH
*     Total Capacitance     C = 12.3218005548298   fF
*
Ready

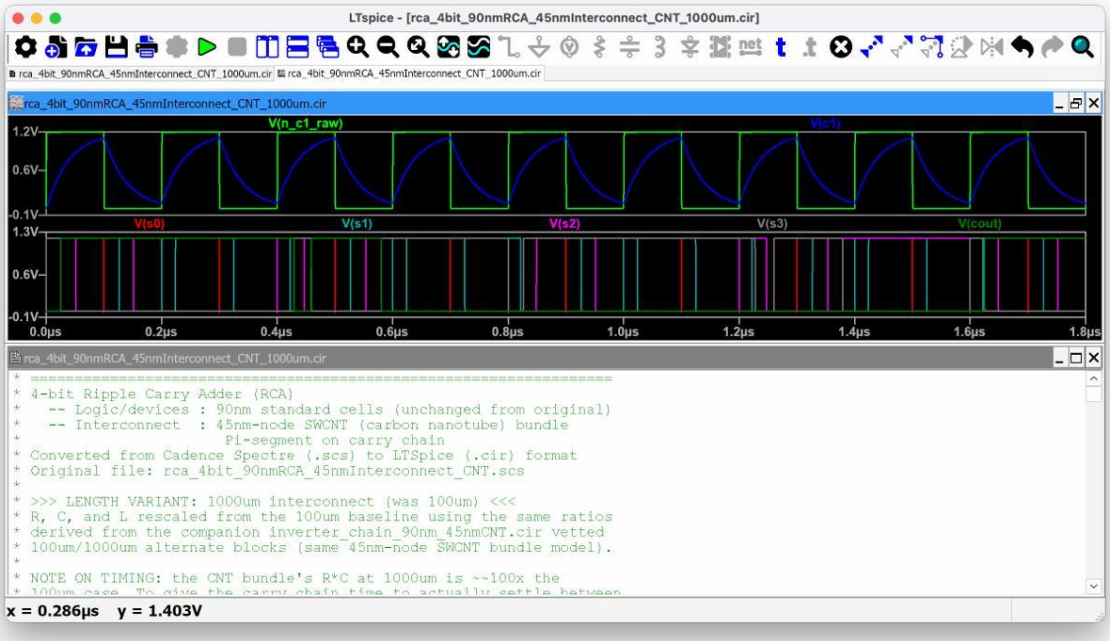

LTspice - [inverter_chain_90nm_14nmCu_500um.cir]
inverter_chain_90nm_14nmCu_500um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
4ns
8ns
12ns
16ns
20ns
24ns
28ns
32ns
36ns
40ns
inverter_chain_90nm_14nmCu_500um.cir
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 14nm-node Copper Interconnect
* Length variant: 500um (companion to the 100um
* inverter_chain_90nm_14nmCu.cir file -- same topology, longer wire)
*
* NOTE ON TIMING: R and C both scale ~5x from 100um to 500um, so
* R*C scales ~25x, from ~69ps (100um) to ~1.73ns (500um). Pulse
* width/period/.tran below have been stretched vs. the 100um file
* to give the interconnect time to settle (~5.8x R*C margin).
* ==========
* ----------
* Global parameters
* ----------
x = 33.26ns y = 1.367V


LTspice - [rca_4bit_90nmRCA_14nmInterconnect.cir]
rca_4bit_90nmRCA_14nmInterconnect.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
18ns
rca_4bit_90nmRCA_14nmInterconnect.cir
* ==========
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 14nm-node copper Cu Pi-segment on carry chain
* Companion file to rca_4bit_90nmRCA_45nmInterconnect.cir -- identical
* logic netlist and stimulus, only the carry-chain interconnect node
* has been moved from 45nm down to 14nm.
*
* Structure:
*   4 full adders (FA0..FA3) built hierarchically from static CMOS
*   gates (inv, nand2, nor2, and2, or2, xor2), all on 90nm devices.
*
*   Copper interconnect is inserted on the INTERNAL CARRY CHAIN
*   (FA0.cout -> FA1.cin -> FA2.cin -> FA3.cin) with 14nm-node Cu
x = 18.00ns y = 0.971V

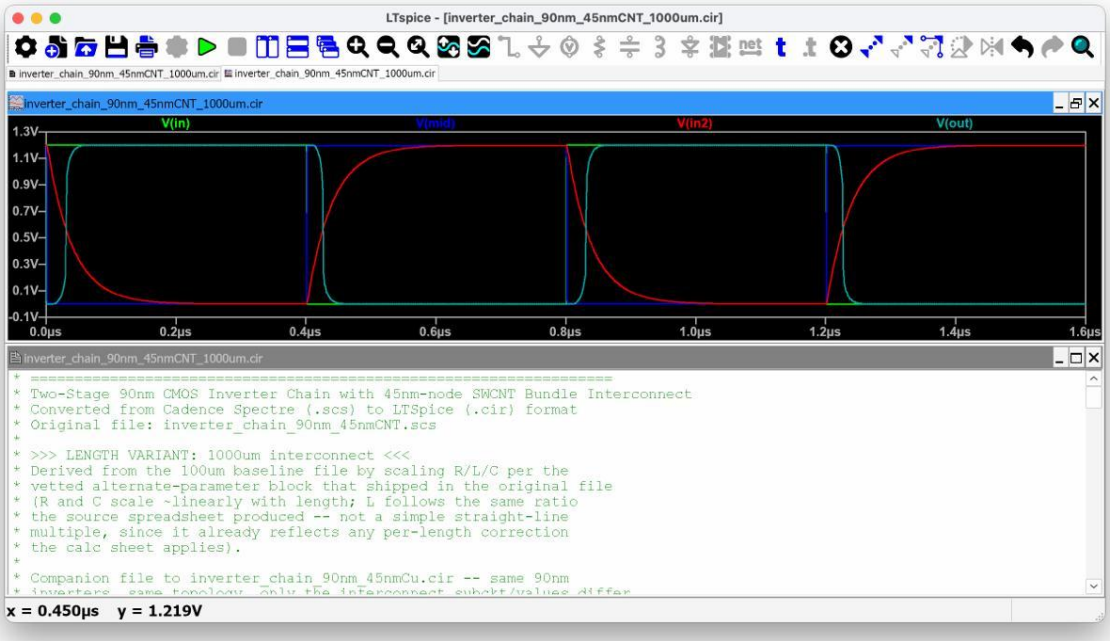

LTspice - [rca_4bit_90nmRCA_14nmInterconnect_CNT.cir]
rca_4bit_90nmRCA_14nmInterconnect_CNT.cir
V(n_c1_raw)
V(c1)
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.2V
0.6V
-0.1V
1.3V
0.6V
-0.1V
0ns
4ns
8ns
12ns
16ns
20ns
24ns
28ns
32ns
36ns
40ns
* ====================================================================
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 14nm-node SWCNT (carbon nanotube) bundle
*                      Pi-segment on carry chain
* Companion file to rca_4bit_90nmRCA_14nmInterconnect.cir -- identical
* logic netlist and stimulus, only the carry-chain interconnect has
* been swapped from Cu to a 14nm-node SWCNT bundle, for a like-for-
* like Cu-vs-CNT comparison at this node.
*
* Structure:
*   4 full adders (FA0..FA3) built hierarchically from static CMOS
*   gates (inv, nand2, nor2, and2, or2, xor2), all on 90nm devices
*   -- identical logic netlist to the Cu companion file.
*
Ready

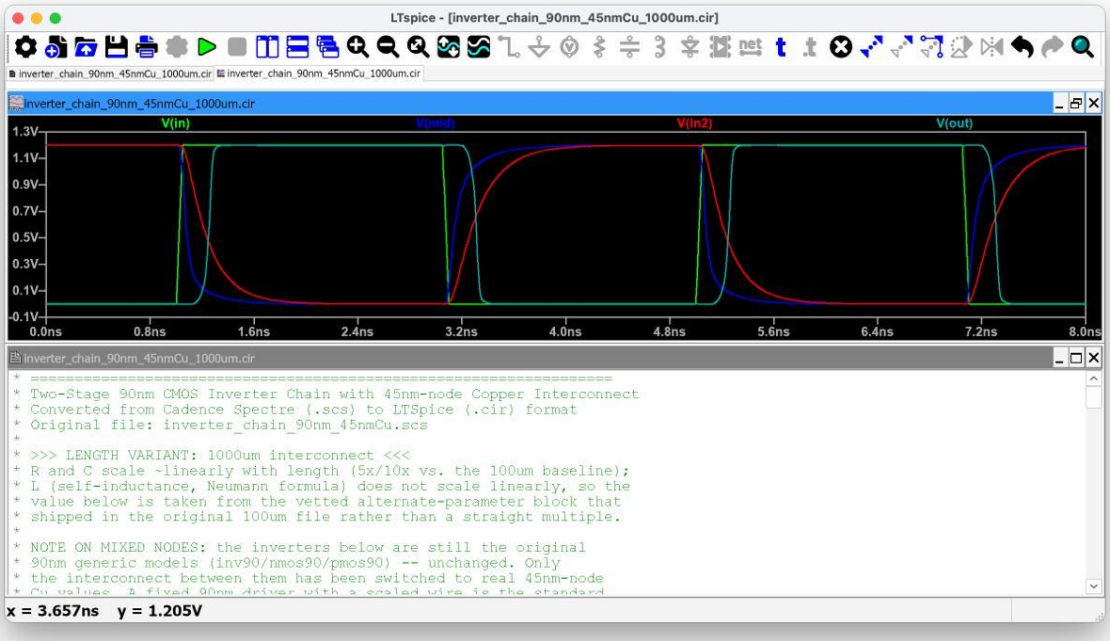

LTspice - [inverter_chain_90nm_14nmCu.cir]
inverter_chain_90nm_14nmCu.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0ns
0.8ns
1.6ns
2.4ns
3.2ns
4.0ns
4.8ns
5.6ns
6.4ns
7.2ns
8.0ns
* ====================================================================
* Two-Stage 90nm CMOS Inverter Chain with 14nm-node Copper Interconnect
* Companion file to inverter_chain_90nm_45nmCu.cir / _45nmCNT.cir --
* same 90nm inverters, same topology, only the interconnect node has
* been moved from 45nm down to 14nm so the two can be compared.
*
* NOTE ON MIXED NODES: the inverters below are still the original
* 90nm generic models (inv90/nmos90/pmos90) -- unchanged. Only
* the interconnect between them has been switched to real 14nm-node
* Cu values. A fixed 90nm driver with a scaled wire is the standard
* way to isolate the interconnect's own contribution to delay.
* ====================================================================
* --------------------------------------------------------------------
* Global parameters
x = 5.819ns  y = 1.304V

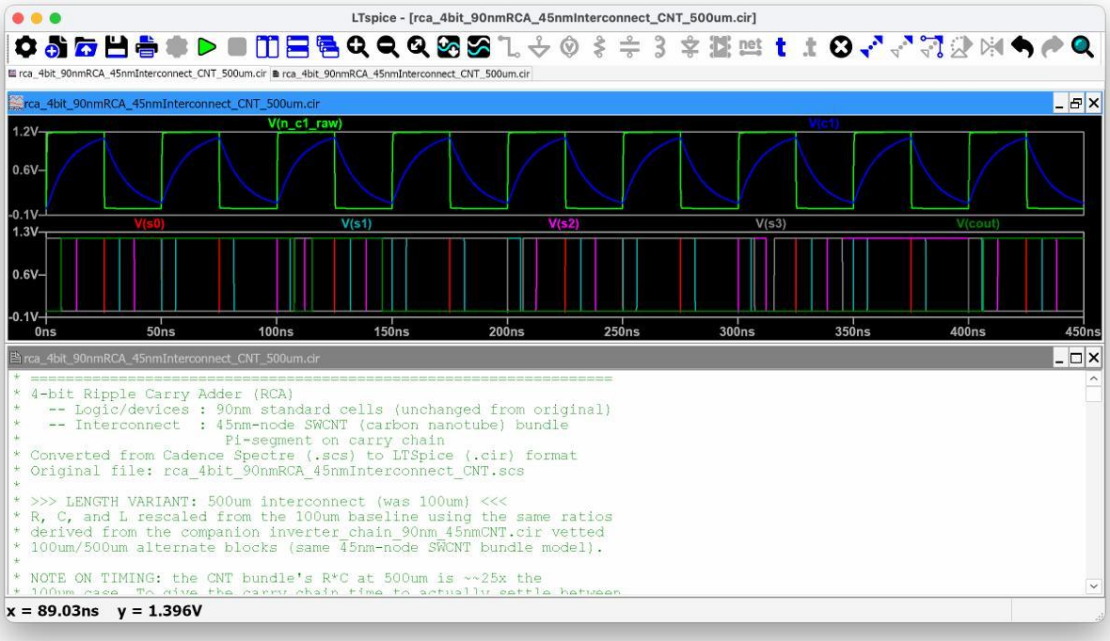

LTspice - [inverter_chain_90nm_14nmCNT.cir]
inverter_chain_90nm_14nmCNT.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
6ns
12ns
18ns
24ns
30ns
36ns
42ns
48ns
54ns
60ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 14nm-node SWCNT Bundle Interconnect
* Companion file to inverter_chain_90nm_14nmCu.cir -- same 90nm
* inverters, same topology, only the interconnect subckt/values differ
* so the two can be run side by side for a direct Cu vs CNT comparison
* at the 14nm node.
*
* NOTE ON TIMING: the CNT bundle's own R*C is ~2.83ns at 100um (vs
* ~69ps for Cu at the same length, and vs ~764ps for the 45nm-node
* CNT bundle), so the pulse period/width and .tran stop time were
* stretched out further than the 45nm CNT file (by roughly the same
* ~3.7x ratio as the R*C increase) to give the output enough time to
* actually settle.
* ==========
x = 59.12ns y = 0.416V

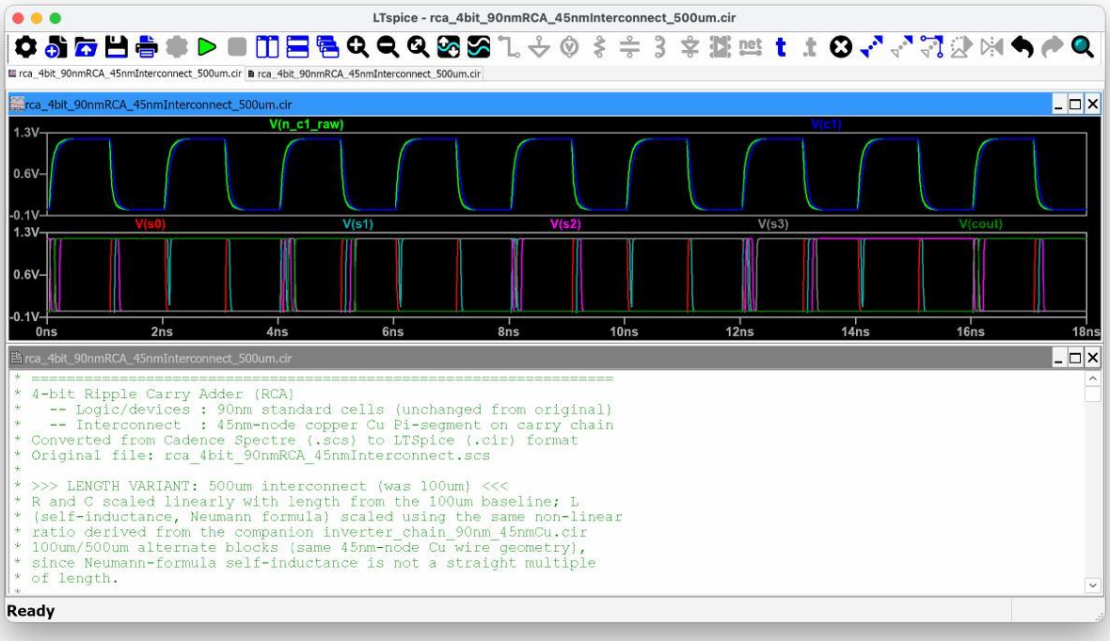

LTspice - [rca_4bit_90nmRCA_45nmInterconnect_CNT_1000um.cir]
rca_4bit_90nmRCA_45nmInterconnect_CNT_1000um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0.0µs
0.2µs
0.4µs
0.6µs
0.8µs
1.0µs
1.2µs
1.4µs
1.6µs
1.8µs
* ==========
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 45nm-node SWCNT (carbon nanotube) bundle
*                      Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_45nmInterconnect_CNT.scs
*
* >>> LENGTH VARIANT: 1000um interconnect (was 100um) <<<
* R, C, and L rescaled from the 100um baseline using the same ratios
* derived from the companion inverter_chain_90nm_45nmCNT.cir vetted
* 100um/1000um alternate blocks (same 45nm-node SWCNT bundle model).
*
* NOTE ON TIMING: the CNT bundle's R*C at 1000um is ~100x the
x = 0.286µs y = 1.403V

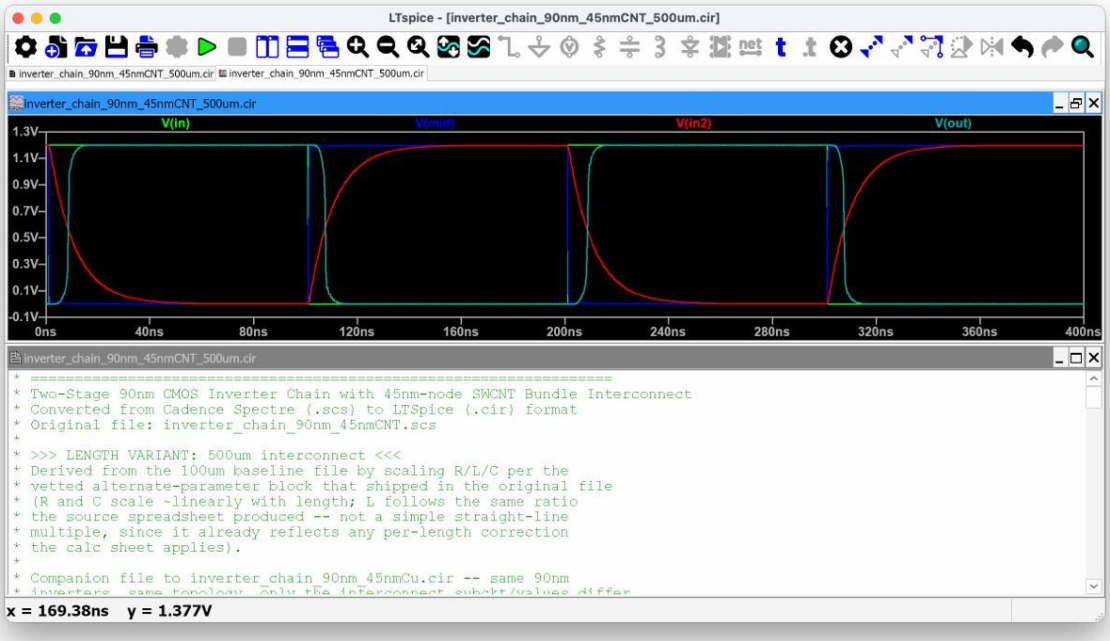

LTspice - [rca_4bit_90nmRCA_45nmInterconnect_1000um.cir]
rca_4bit_90nmRCA_45nmInterconnect_1000um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
18ns
rca_4bit_90nmRCA_45nmInterconnect_1000um.cir
* ==========
* 4-bit Ripple Carry Adder (RCA)
* -- Logic/devices : 90nm standard cells (unchanged from original)
* -- Interconnect : 45nm-node copper Cu Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_45nmInterconnect.scs
*
* >>> LENGTH VARIANT: 1000um interconnect (was 100um) <<<
* R and C scaled linearly with length from the 100um baseline; L
* (self-inductance, Neumann formula) scaled using the same non-linear
* ratio derived from the companion inverter_chain_90nm_45nmCu.cir
* 100um/1000um alternate blocks (same 45nm-node Cu wire geometry),
* since Neumann-formula self-inductance is not a straight multiple
* of length.
*
x = 6.47ns y = 1.004V

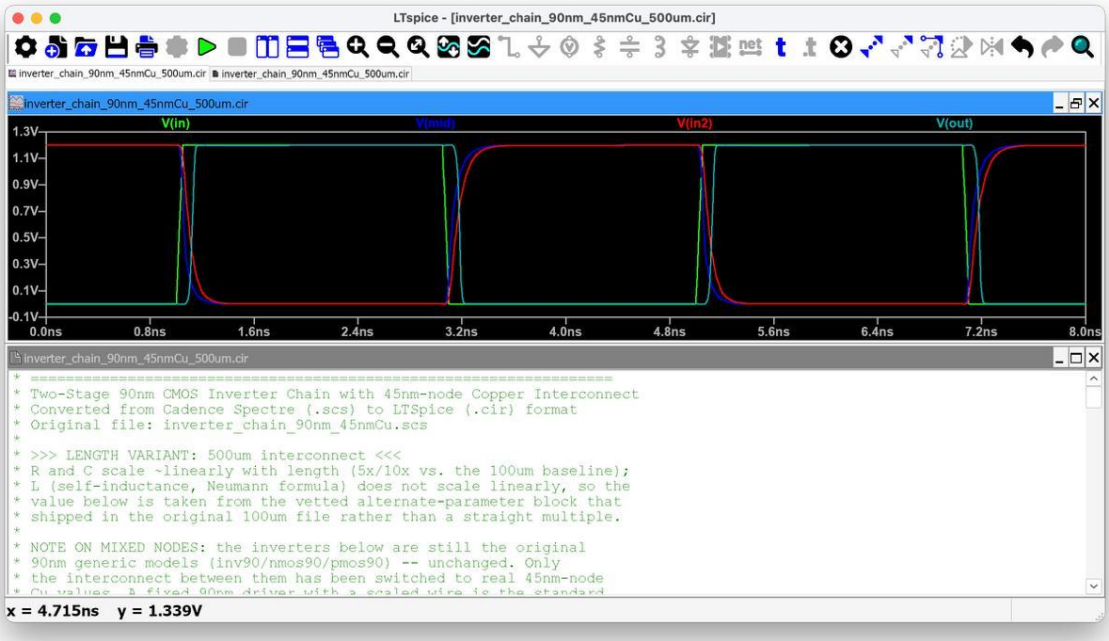

LTspice - [inverter_chain_90nm_45nmCNT_1000um.cir]
inverter_chain_90nm_45nmCNT_1000um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0µs
0.2µs
0.4µs
0.6µs
0.8µs
1.0µs
1.2µs
1.4µs
1.6µs
inverter_chain_90nm_45nmCNT_1000um.cir
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 45nm-node SWCNT Bundle Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_45nmCNT.scs
*
* >>> LENGTH VARIANT: 1000um interconnect <<<
* Derived from the 100um baseline file by scaling R/L/C per the
* vetted alternate-parameter block that shipped in the original file
* (R and C scale ~linearly with length; L follows the same ratio
* the source spreadsheet produced -- not a simple straight-line
* multiple, since it already reflects any per-length correction
* the calc sheet applies).
*
* Companion file to inverter_chain_90nm_45nmCu.cir -- same 90nm
* inverters, same topology, only the interconnect subckt values differ
x = 0.450µs y = 1.219V

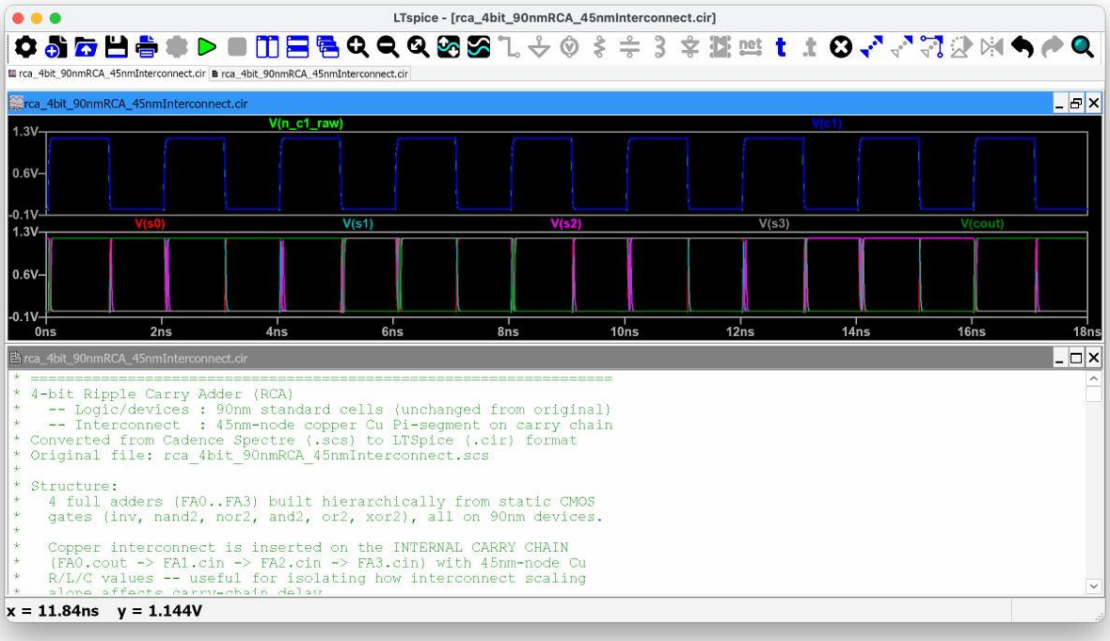

LTspice - [inverter_chain_90nm_45nmCu_1000um.cir]
inverter_chain_90nm_45nmCu_1000um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0ns
0.8ns
1.6ns
2.4ns
3.2ns
4.0ns
4.8ns
5.6ns
6.4ns
7.2ns
8.0ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 45nm-node Copper Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_45nmCu.scs
*
* >>> LENGTH VARIANT: 1000um interconnect <<<
* R and C scale ~linearly with length (5x/10x vs. the 100um baseline);
* L (self-inductance, Neumann formula) does not scale linearly, so the
* value below is taken from the vetted alternate-parameter block that
* shipped in the original 100um file rather than a straight multiple.
*
* NOTE ON MIXED NODES: the inverters below are still the original
* 90nm generic models (inv90/nmos90/pmos90) -- unchanged. Only
* the interconnect between them has been switched to real 45nm-node
x = 3.657ns y = 1.205V

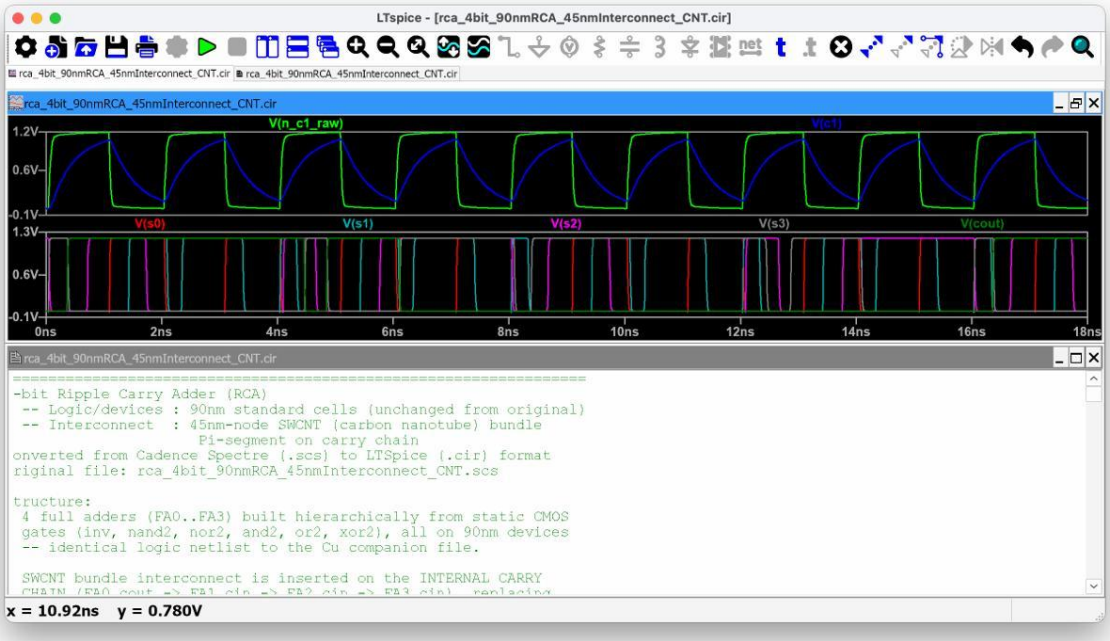

LTspice - [rca_4bit_90nmRCA_45nmInterconnect_CNT_500um.cir]
rca_4bit_90nmRCA_45nmInterconnect_CNT_500um.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
50ns
100ns
150ns
200ns
250ns
300ns
350ns
400ns
450ns
* ==========
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 45nm-node SWCNT (carbon nanotube) bundle
*                      Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_45nmInterconnect_CNT.scs
*
* >>> LENGTH VARIANT: 500um interconnect (was 100um) <<<
* R, C, and L rescaled from the 100um baseline using the same ratios
* derived from the companion inverter_chain_90nm_45nmCNT.cir vetted
* 100um/500um alternate blocks (same 45nm-node SWCNT bundle model).
*
* NOTE ON TIMING: the CNT bundle's R*C at 500um is ~25x the
x = 89.03ns y = 1.396V

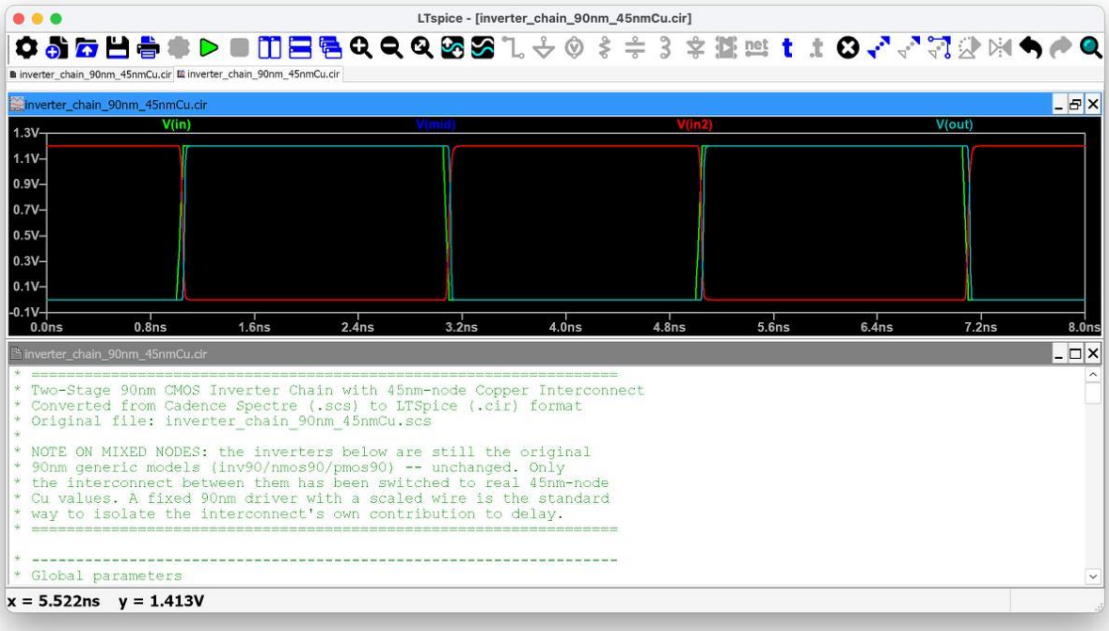

LTspice - rca_4bit_90nmRCA_45nmInterconnect_500um.cir
rca_4bit_90nmRCA_45nmInterconnect_500um.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
18ns
* ================================================================
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 45nm-node copper Cu Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_45nmInterconnect.scs
*
* >>> LENGTH VARIANT: 500um interconnect (was 100um) <<<
* R and C scaled linearly with length from the 100um baseline; L
* (self-inductance, Neumann formula) scaled using the same non-linear
* ratio derived from the companion inverter_chain_90nm_45nmCu.cir
* 100um/500um alternate blocks (same 45nm-node Cu wire geometry),
* since Neumann-formula self-inductance is not a straight multiple
* of length.
*
Ready

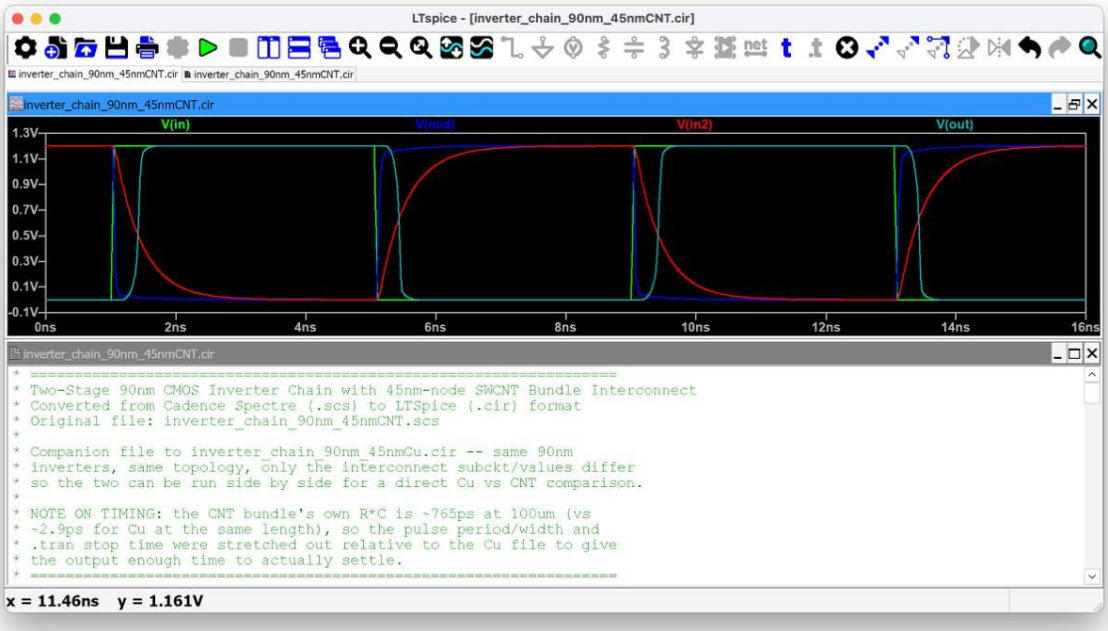

LTspice - [inverter_chain_90nm_45nmCNT_500um.cir]
inverter_chain_90nm_45nmCNT_500um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
40ns
80ns
120ns
160ns
200ns
240ns
280ns
320ns
360ns
400ns
* ================================================================
* Two-Stage 90nm CMOS Inverter Chain with 45nm-node SWCNT Bundle Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_45nmCNT.scs
*
* >>> LENGTH VARIANT: 500um interconnect <<<
* Derived from the 100um baseline file by scaling R/L/C per the
* vetted alternate-parameter block that shipped in the original file
* (R and C scale ~linearly with length; L follows the same ratio
* the source spreadsheet produced -- not a simple straight-line
* multiple, since it already reflects any per-length correction
* the calc sheet applies).
*
* Companion file to inverter_chain_90nm_45nmCu.cir -- same 90nm
x = 169.38ns  y = 1.377V

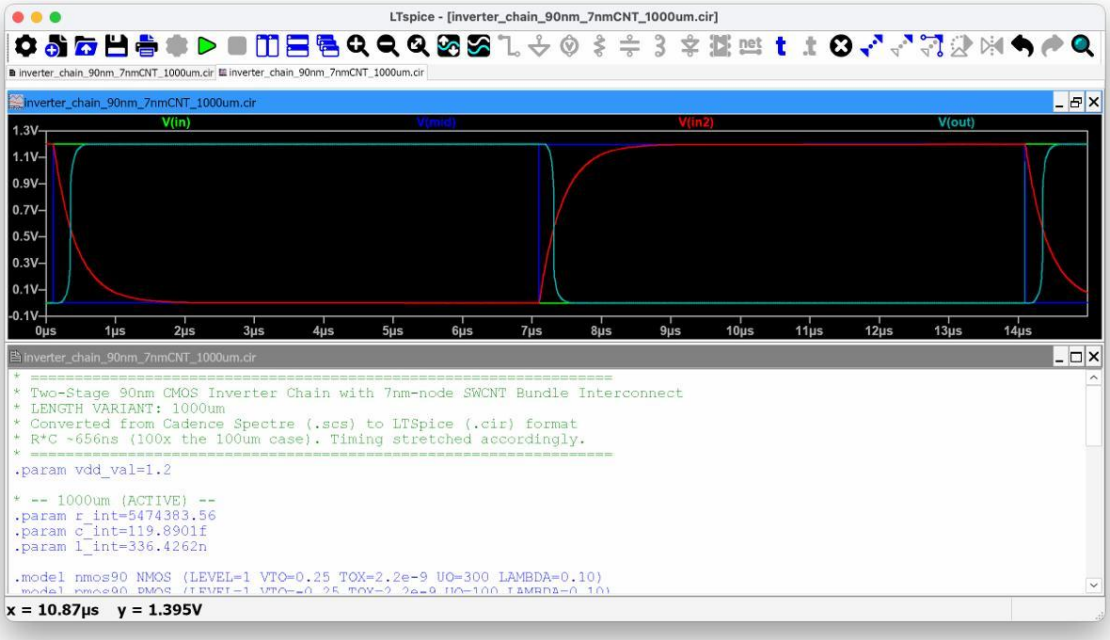

LTspice - [inverter_chain_90nm_45nmCu_500um.cir]
inverter_chain_90nm_45nmCu_500um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0ns
0.8ns
1.6ns
2.4ns
3.2ns
4.0ns
4.8ns
5.6ns
6.4ns
7.2ns
8.0ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 45nm-node Copper Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_45nmCu.scs
*
* >>> LENGTH VARIANT: 500um interconnect <<<
* R and C scale ~linearly with length (5x/10x vs. the 100um baseline);
* L (self-inductance, Neumann formula) does not scale linearly, so the
* value below is taken from the vetted alternate-parameter block that
* shipped in the original 100um file rather than a straight multiple.
*
* NOTE ON MIXED NODES: the inverters below are still the original
* 90nm generic models (inv90/nmos90/pmos90) -- unchanged. Only
* the interconnect between them has been switched to real 45nm-node
x = 4.715ns y = 1.339V


LTspice - [rca_4bit_90nmRCA_45nmInterconnect.cir]
rca_4bit_90nmRCA_45nmInterconnect.cir
V(n_c1_raw)
V(c1)
1.3V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
18ns
* ==========
* 4-bit Ripple Carry Adder (RCA)
*   -- Logic/devices : 90nm standard cells (unchanged from original)
*   -- Interconnect  : 45nm-node copper Cu Pi-segment on carry chain
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: rca_4bit_90nmRCA_45nmInterconnect.scs
*
* Structure:
*   4 full adders (FA0..FA3) built hierarchically from static CMOS
*   gates (inv, nand2, nor2, and2, or2, xor2), all on 90nm devices.
*
*   Copper interconnect is inserted on the INTERNAL CARRY CHAIN
*   (FA0.cout -> FA1.cin -> FA2.cin -> FA3.cin) with 45nm-node Cu
*   R/L/C values -- useful for isolating how interconnect scaling
x = 11.84ns y = 1.144V


LTspice - [rca_4bit_90nmRCA_45nmInterconnect_CNT.cir]
rca_4bit_90nmRCA_45nmInterconnect_CNT.cir
V(n_c1_raw)
V(c1)
1.2V
0.6V
-0.1V
V(s0)
V(s1)
V(s2)
V(s3)
V(cout)
1.3V
0.6V
-0.1V
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
18ns
rca_4bit_90nmRCA_45nmInterconnect_CNT.cir
-bit Ripple Carry Adder (RCA)
-- Logic/devices : 90nm standard cells (unchanged from original)
-- Interconnect : 45nm-node SWCNT (carbon nanotube) bundle
Pi-segment on carry chain
onverted from Cadence Spectre (.scs) to LTSpice (.cir) format
riginal file: rca_4bit_90nmRCA_45nmInterconnect_CNT.scs
tructure:
4 full adders (FA0..FA3) built hierarchically from static CMOS
gates (inv, nand2, nor2, and2, or2, xor2), all on 90nm devices
-- identical logic netlist to the Cu companion file.
SWCNT bundle interconnect is inserted on the INTERNAL CARRY
x = 10.92ns y = 0.780V


LTspice - [inverter_chain_90nm_45nmCu.cir]
inverter_chain_90nm_45nmCu.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0.0ns
0.8ns
1.6ns
2.4ns
3.2ns
4.0ns
4.8ns
5.6ns
6.4ns
7.2ns
8.0ns
inverter_chain_90nm_45nmCu.cir
* ======================================================================
* Two-Stage 90nm CMOS Inverter Chain with 45nm-node Copper Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_45nmCu.scs
*
* NOTE ON MIXED NODES: the inverters below are still the original
* 90nm generic models (inv90/nmos90/pmos90) -- unchanged. Only
* the interconnect between them has been switched to real 45nm-node
* Cu values. A fixed 90nm driver with a scaled wire is the standard
* way to isolate the interconnect's own contribution to delay.
* ======================================================================
* ----------------------------------------------------------------------
* Global parameters
x = 5.522ns y = 1.413V


LTspice - [inverter_chain_90nm_45nmCNT.cir]
inverter_chain_90nm_45nmCNT.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0ns
2ns
4ns
6ns
8ns
10ns
12ns
14ns
16ns
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 45nm-node SWCNT Bundle Interconnect
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* Original file: inverter_chain_90nm_45nmCNT.scs
*
* Companion file to inverter_chain_90nm_45nmCu.cir -- same 90nm
* inverters, same topology, only the interconnect subckt/values differ
* so the two can be run side by side for a direct Cu vs CNT comparison.
*
* NOTE ON TIMING: the CNT bundle's own R*C is ~765ps at 100um (vs
* ~2.9ps for Cu at the same length), so the pulse period/width and
* .tran stop time were stretched out relative to the Cu file to give
* the output enough time to actually settle.
* ==========
x = 11.46ns y = 1.161V


LTspice - [inverter_chain_90nm_7nmCNT_1000um.cir]
inverter_chain_90nm_7nmCNT_1000um.cir
V(in)
V(mid)
V(in2)
V(out)
1.3V
1.1V
0.9V
0.7V
0.5V
0.3V
0.1V
-0.1V
0µs
1µs
2µs
3µs
4µs
5µs
6µs
7µs
8µs
9µs
10µs
11µs
12µs
13µs
14µs
* ==========
* Two-Stage 90nm CMOS Inverter Chain with 7nm-node SWCNT Bundle Interconnect
* LENGTH VARIANT: 1000um
* Converted from Cadence Spectre (.scs) to LTSpice (.cir) format
* R*C ~656ns (100x the 100um case). Timing stretched accordingly.
* ==========
.param vdd_val=1.2
* -- 1000um (ACTIVE) --
.param r_int=5474383.56
.param c_int=119.8901f
.param l_int=336.4262n
.model nmos90 NMOS (LEVEL=1 VTO=0.25 TOX=2.2e-9 UO=300 LAMBDA=0.10)
x = 10.87µs y = 1.395V